

NIKITA BHANSALI

☎ +91-8585804977 ✉ nbhansali2006@gmail.com  [nikitabhansali](#)  [nikitaaaa123](#)  [nikita-portfolio-beige.vercel.app](#)

EDUCATION

Vellore Institute of Technology

September 2023 – October 2027

Bachelor of Technology in Electronics and Communication Engineering; CGPA:8.48

SKILL SET

Embedded Systems & Hardware: ESP32/ESP32-CAM, Arduino IDE, Sensor Interfacing (DHT11, PIR)

RF & Signal Processing: Software Defined Radio (RTL-SDR, HackRF, ADALM-PLUTO), DragonOS

Programming Languages: Python, Embedded C, Java

Databases: MySQL, MongoDB

Frameworks: MERN Stack, GitHub

INTERNSHIP

MPOnline (Government of MP) | *AI/ML & Advanced Software Engineering Internship*

May 2026 – July 2026

-Built ML models (regression, classification, decision trees, random forest, SVM, clustering, PCA) using Python, NumPy, Pandas, and Scikit-learn, and developed neural networks with TensorFlow/PyTorch for computer vision and NLP.

-Practiced full-stack SDLC: DBMS, SQL, OOP, Git/GitHub, and CRUD app development, followed by requirement analysis, system design, backend development, testing, and cloud deployment for a capstone project.

PROJECTS

Anti-Drone System using SDR | *SDR, RF Systems, Embedded Design*

August 2026 – March 2027

-Engineered a dual-drone counter-UAV system in which a programmable ESP32 "Enemy Drone" simulates rogue RF intrusions across the 2.4 GHz/5.8 GHz ISM bands for campus perimeter security testing.

-Configured a ground station with an ADALM-PLUTO SDR on DragonOS (GNU Radio, SigDigger) to isolate the enemy drone's RF signature and compute real-time signal bearing and RSSI.

-Established a sub-5ms ESP-NOW ground-to-air relay to guide a Defender ESP32 drone toward the target and trigger localized non-kinetic disruption (Wi-Fi de-authentication) to force a controlled landing.

Home Automation with LumaSync | *IoT, Embedded Systems, ESP32*

January 2025 – April 2025

-IoT-based home automation and security system regulating room temperature, detecting motion, and capturing images of entrants, controllable via a local web dashboard and Google Assistant.

-Assembled the system using an ESP32/ESP32-CAM, a DHT11 sensor, a PIR sensor, and a 4-channel relay module; developed a real-time dashboard (HTML/CSS/JS) and integrated Google Assistant via IFTTT.

-Devised predictive, trend-based temperature control logic (dynamic hysteresis) to reduce overshoot and improve system responsiveness.

Smart Surveillance for Public Transport | *Computer Vision, Embedded AI*

September 2024 – January 2025

-Real-time monitoring system for public transport to detect anomalies and enhance passenger safety.

-Implemented OpenCV and YOLO for real-time object detection and video analysis on embedded/edge hardware.

-Fine-tuned anomaly-detection models and optimized YOLO inference for real-time performance on live feeds.

CERTIFICATIONS

Health Hackathon 2026 – Finalist

[\[tinyurl.com/HealthHack26\]](#)

Health Hackathon 2025 – Finalist

[\[tinyurl.com/HealthHack25\]](#)

EXTRACURRICULAR

Languages: Proficient in English, Hindi; learning Japanese and German (300+ day Duolingo streak).

[\[tinyurl.com/300Duolingo\]](#)

Leadership: Student Partner at IEEE and VITronix Club, organizing technical workshops for 1000+ students.

National Service Scheme: Served as Hospitality Lead for the NSS Unit Camp and led the Tree Plantation Drive team.

HOBBIES

Anchoring, Dancing, Event Management, Photography, Public Speaking.